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Specialization: *Full scan of the NYPD historic data*

Business Focus: *Security industry*

Tool: *python, databrocks, azure*

End to End Data Engineering Project (NYPD) using Azure

Project Learning Opportunities

understanding of azure cloud platform, using pyspark within databricks

Tools and Technology to be Used

python™

databricks

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Case Study Overview

Introduction to the Business

The New York City Police Department (NYPD) is responsible for law enforcement across New York's five boroughs. It handles hundreds of thousands of arrests annually. Using a comprehensive historic arrest dataset, you can explore patterns, trends, and disparities in policing over a 15-year span—from 2006 through 2020

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Case Study Overview

Problem Statement

How have arrest patterns—by type of offense, location, demographics, and over time—changed in New York City from 2006 to 2020? Are there identifiable disparities across neighborhoods or demographic groups, and what policy or external factors influenced these patterns?

Sub-questions:

- What are the temporal trends in arrest volume (year-to-year changes, seasonality)?
- How do arrest types breakdown (e.g. violent vs property vs quality-of-life offenses)?
- Do arrest rates differ significantly by borough or precinct?
- Are there demographic disparities in arrests by race, age group, or gender?

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Rationale for the Project

(What is the Importance of the project to the business)

1.

Public Transparency & Accountability:

Citizens and policymakers seek insights into how arrest practices have evolved over time.

2.

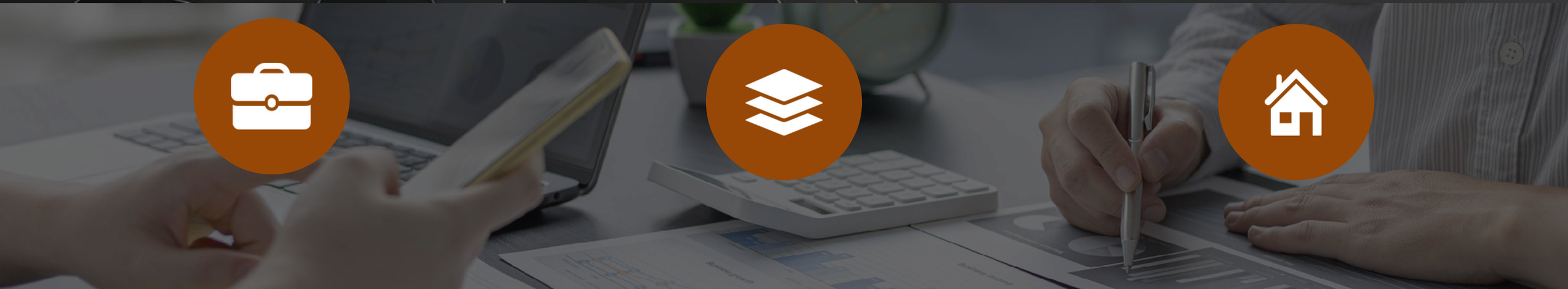
Equity Analysis:

With suspect demographic data, the study can reveal whether arrests differ by race, age, sex, or neighborhood—highlighting potential bias or disproportionality.

3.

Public Safety & Resource Planning:

Analyzing trends helps allocate resources and tailor interventions in high-arrest zones.



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Case Study Objectives

(What participants will be able to do at the end of the case study)

Build a Scalable Data Pipeline

- Design and implement an automated pipeline to ingest and process the large historical NYPD dataset from Kaggle.
- Ensure the pipeline supports both batch ingestion for historical data and incremental updates for future refreshes.

2. Implement Robust Data Storage

- Store raw and processed data in an optimized structure for analytical queries.
- Use data lake + data warehouse architecture (e.g., AWS S3/ADLS + Snowflake/BigQuery).
- Apply partitioning by date or borough to improve query performance.

3. Ensure Data Quality & Consistency

- Clean and standardize data (handle missing values, normalize offense types, correct data types).
- Implement data validation checks (schema validation, null checks, range checks).
- Track and log data lineage for transparency.

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Data Description

ARREST_KEY: Randomly generated persistent ID for each arrest
ARREST_DATE: Exact date of arrest for the reported event
PD_CD: Three digit internal classification code (more granular than Key Code)
PD_DESC: Description of internal classification corresponding with PD code (more granular than Offense Description)
KY_CD: Three digit internal classification code (more general category than PD code)
OFNS_DESC: Description of internal classification corresponding with KY code (more general category than PD description)
LAW_CODE: Law code charges corresponding to the NYS Penal Law, VTL and other various local laws
LAW_CAT_CD: Level of offense: felony, misdemeanor, violation
ARREST_BORO: Borough of arrest. B(Bronx), S(Staten Island), K(Brooklyn), M(Manhattan), Q(Queens)
ARREST_PRECINCT: Precinct where the arrest occurred
JURISDICTION_CODE: Jurisdiction responsible for arrest. Jurisdiction codes 0(Patrol), 1(Transit) and 2(Housing) represent NYPD whilst codes 3 and more represent non NYPD jurisdictions
AGE_GROUP: Perpetrator's age within a category
PERP_SEX: Perpetrator's sex description
PERP_RACE: Perpetrator's race description
X_COORD_CD: Midblock X-coordinate for New York State Plane Coordinate System, Long Island Zone, NAD 83, units feet (FIPS 3104)
Y_COORD_CD: Midblock Y-coordinate for New York State Plane Coordinate System, Long Island Zone, NAD 83, units feet (FIPS 3104)
Latitude: Latitude coordinate for Global Coordinate System, WGS 1984, decimal degrees (EPSG 4326)
Longitude: Longitude coordinate for Global Coordinate System, WGS 1984, decimal degrees (EPSG 4326)

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Tech Stack

python™

databricks

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Project Workflow

STEP 1 Set Up Data Sources: We would be using NYPD open data csv file

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STEP 2 Create Storage Layers

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STEP 3 Ingestion: Using data factory

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STEP 4 Data Transformation: Using databricks

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STEP 5 Data Warehouse: Azure synapse

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